Cormalent Film-Coated Tablets

Module 3 Quality Documentation — Formulation F-399/A

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement.

Manufacturing Process Overview

The drug product is manufactured by a conventional wet granulation process comprising dispensing, high-shear granulation, fluid-bed drying, milling, blending, compression, and film coating. In-process controls are applied at each unit operation, including loss on drying after fluid-bed drying and blend uniformity prior to compression. Process parameters were established during process performance qualification and are maintained under the site quality system.

Deviations observed during the campaign were investigated and closed with no impact to product quality. No changes to the approved analytical procedures are proposed in this submission. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Hold-time studies established acceptable storage durations for intermediate blends and cores under defined conditions. No changes to the approved analytical procedures are proposed in this submission. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Section 3.2.P.4 — Excipients

For the commercial formulation F-399/A, Aerosil 200 Pharma (Colloidal silicon dioxide, glidant) is used.

Formulation Development Rationale

Aerosil 200 Pharma was selected as the commercial grade of colloidal silicon dioxide on the basis of lower moisture content mitigating hydrolytic degradation of the drug substance.

Colloidal silicon dioxide complies with the requirements of ICH Q3D and FDA Guidance for Industry: Immediate Release Solid Oral Dosage Forms.

Commercial Control Strategy — Excipients

The current approved commercial specification (SPEC-EX-5025 Rev 09) lists Colloidal silicon dioxide as Aerosil 200 Pharma.

Reference Standards

Primary reference standards for the drug substance and specified impurities are qualified against compendial standards where available. Working standards are qualified against the primary standard under an approved protocol and are stored under controlled conditions.

Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. No changes to the approved analytical procedures are proposed in this submission.

Analytical Method Validation Summary

The assay and related-substances procedures employ reversed-phase high performance liquid chromatography with ultraviolet detection. Validation demonstrated acceptable specificity, linearity, accuracy, precision, and robustness. Forced degradation studies confirmed the stability-indicating capability of the related-substances method.

The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. Statistical evaluation of batch release data demonstrates process capability well within specification limits. The control strategy links critical quality attributes to critical process parameters identified during development. Training records for personnel involved in manufacturing and testing are maintained under the site quality system.

Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Statistical evaluation of batch release data demonstrates process capability well within specification limits.

Regional Information

No excipients of human or animal origin are used in the drug product with the exception of lactose monohydrate, for which a TSE/BSE compliance declaration is provided in Module 3.2.A.2. Certificates of suitability are maintained where applicable.

Deviations observed during the campaign were investigated and closed with no impact to product quality. The finished product is stored and distributed under controlled ambient conditions consistent with the approved label storage statement. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable.

Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Environmental monitoring of the manufacturing areas is performed per the site monitoring program, with alert and action limits established from historical data. Statistical evaluation of batch release data demonstrates process capability well within specification limits. Change control procedures ensure that any modification to materials, methods, or equipment is assessed for regulatory impact prior to implementation. Analytical results are trended electronically, and out-of-trend results trigger a documented laboratory investigation.

Stability Summary

Primary stability studies were conducted under ICH long-term (25°C/60% RH) and accelerated (40°C/75% RH) conditions on three registration batches. All quality attributes remained within acceptance criteria through the tested intervals. Photostability testing per ICH Q1B confirmed that the product is not photolabile in the commercial pack.

Statistical evaluation of batch release data demonstrates process capability well within specification limits. Training records for personnel involved in manufacturing and testing are maintained under the site quality system. Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Comparability of pilot- and commercial-scale batches was demonstrated with respect to dissolution and impurity profiles. Data supporting these conclusions are maintained at the manufacturing site and are available for inspection. Cleaning verification swab results were below the health-based exposure limits derived for the product.

Annual product quality reviews evaluate batch trends, deviations, complaints, and stability data to confirm the continued state of control. Risk assessments were performed in accordance with ICH Q9 principles, and residual risks were judged acceptable. Supplier qualification includes periodic audits, review of certificates of analysis, and trending of incoming material test results. The control strategy links critical quality attributes to critical process parameters identified during development.